MASTER'S COMPREHENSIVE EXAMINATION

Part I **Theory (90 minutes)** **March 8, 2008**

Instructions: Attempt any three out of the following four questions. Mention which three you would like to have graded. Open book and open notes.

1. Let $X_1 \sim N(u, \sigma^2)$ $X_2 \sim N(u, \sigma^2+\tau^2)$ and $X_3 \sim N(u, \tau^2)$ with X_1, X_2 and X_3 independent, where σ^2 and τ^2 represent two components of variance

a. Find the MSE of the following estimators of u:

$$\tilde{u} = \frac{X_1 + 3X_2 + 2X_3}{6}$$

$$\bar{u} = \frac{X_1 + X_2 + X_3}{3}$$

b. For what values of σ^2 and τ^2 is $\tilde{u}$ a better estimator of u than $\bar{u}$ is?

2. The distribution of the time to when an oil drilling bit breaks down in years is:

Bit-A $f(x) = 2e^{-2x}$ for $x > 0$
Bit-B $g(y) = 3e^{-3y}$ for $y > 0$

A company buys one Bit-A drill and one Bit-B drill and immediately begins to use both drills. It costs \$8,000 for Bit-A and \$10,000 for Bit-B and each of these drill bits will need to be replaced immediately with the same type of bit if they break down once within one year.

A. What is the probability that either Bit A or Bit B will break down within one year. No credit if you do not explain your answer.

B. Which drill Bit, Bit-A or Bit-B is more cost effective in terms of years of life per dollar spent, no credit if you do not justify your answer?

C. What is the probability that the Bit-A drill just purchased will last longer than the Bit-B drill just purchased, assume that the time to breakdown of different drill bits is independent?

3. The number of meteorites n that hits a planet in a month is a random variable rhar follows the "Busch-Livingston (BL) distribution with:

$$f(x) = \frac{e^a}{(1+e^a)^{x+1}} \quad x = 0, 1, 2, \ldots..$$

A. If x=0, what is the likelihood ratio for a=2 Vs a=1

B. Find the rejection region for n to test H_o: a = 0, Vs H_a: a >0 with size 0.0625

C. Find the power of the test in b to reject H_o when a=ln(0.5) (the natural log of 0.5)

4. Weights of Rocks from the "Blue Sea" are distributed normally with mean 25 kg and variance 25. A scientist needs to find two rocks for different types of analyses.

<u>Analysis I</u> The rock must weigh less than 18 kg
<u>Analysis II</u> The rock must weigh more than 30 kg.

A. If one rock is randomly chosen, what is the probability that this rock will be useful for at least one of the analyses (Analysis I or Analysis II)?

B. It the one rock chosen is not useful for Analysis I, what is the probability that it is useful for Analysis II?

C. If three rocks are randomly chosen, what is the probability that among these three rocks there will be <u>at least one rock which is useful for Analysis I and also at least one rock which is useful for Analysis II</u>? Assume the weights of all three rocks chosen are independent.